EE200: Signals, Systems and Networks

Course Project — Q1A & Q1B

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Q1A. Frequency Forensics — "The Ghost Signal"

Objective

The objective is to identify and remove periodic interference from a corrupted image using frequency-domain filtering and recover the hidden message.

Approach

An image $I(x, y)$ is a two-dimensional discrete signal.
For an $M \times N$ image, the 2D Discrete Fourier Transform (DFT) and its inverse are

$$F(u,v) = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1} I(x,y)\, e^{-j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}, \qquad I(x,y) = \frac{1}{MN}\sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F(u,v)\, e^{j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}. \tag{1}$$

Procedure to find the hidden message is:

1. We first computed the 2D Fourier transform of the image using `fft2` and visualise the magnitude spectrum on both linear and dB scales,

   $$dB = 20\log_{10}(|F| + \varepsilon).$$

   The linear-scale spectrum is dominated by the DC component, which makes the interference peaks difficult to identify. Using a dB scale compresses the dynamic range and makes these peaks clearly visible.

2. We will apply `fftshift` so that the DC component is shifted to the centre of the spectrum. This makes the frequency distribution symmetric and easier to visualize the interference peaks.

3. Since this is a real image so it follows:

   $$F(-u, -v) = F^*(u, v)$$

   Interference peak will be symmetric about the new position of DC component, so we make a central disk about the DC centre to filter out the corrupted periodic part.

4. Now we build a notch filter $H(u, v)$: equal to 1 everywhere, except small disks of radius r_{notch} centred on each detected peak, where it drops to 0.

5. Now we will apply the transformation

$$G(u, v) = F(u, v) \cdot H(u, v)$$

`ifftshift`, inverse-FFT, and keep only the real part to get the recovered image $\hat{I}(x, y)$ back.

We will use percentile as a parameter for the filter to remove the frequency and see it's effects on interfernce. Higher perfenctile means removing frequencies but lesser in quantities. Higher percentile preserves the image which removing the strongest noise frequency.

Results

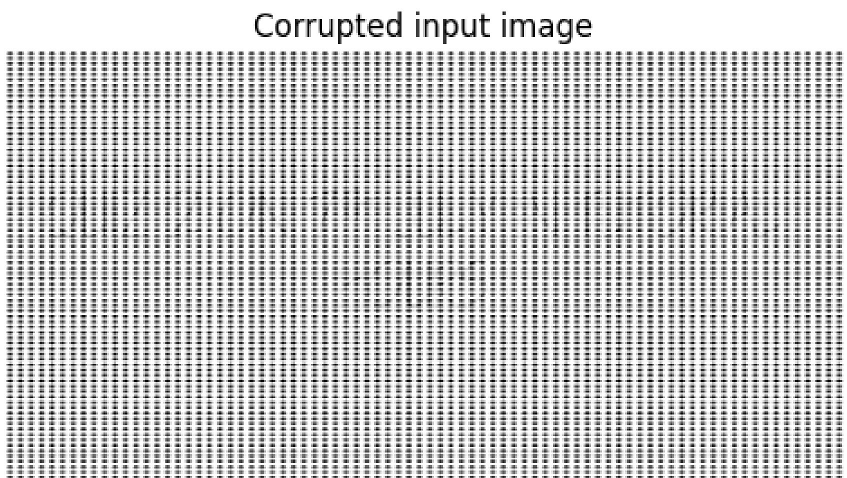


Figure 1: The corrupted input image — a dense grid of dots covering the whole picture.

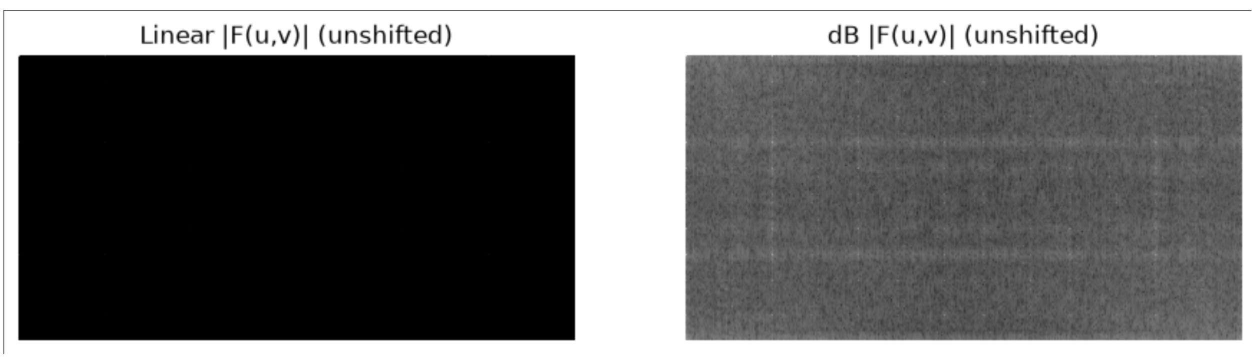


Figure 2: Magnitude spectrum on a linear scale (left) versus a dB scale (right), before shifting — DC sits at the corner $(0, 0)$ in both.

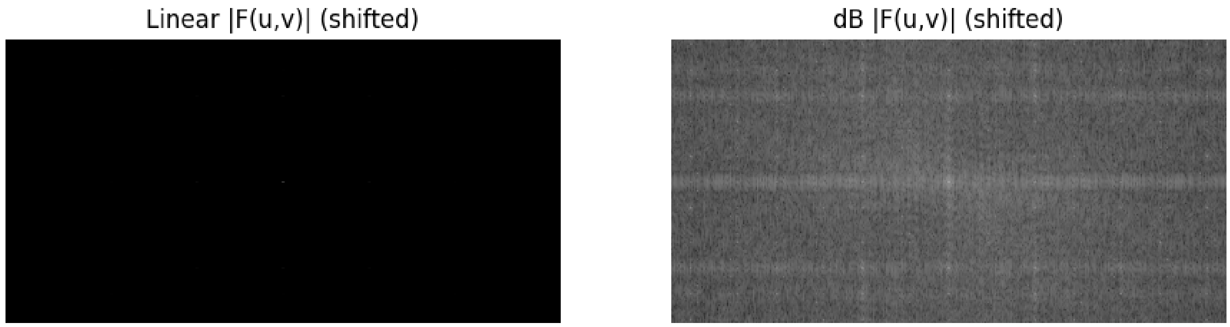


Figure 3: The same spectrum after `fftshift`, linear scale (left) and dB scale (right). DC is now centred, which makes the symmetry of the noise much easier to read.

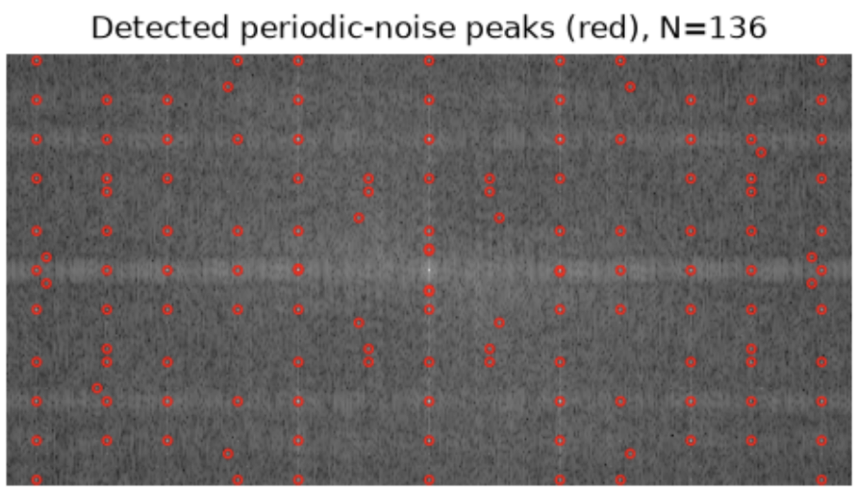


Figure 4: Detected periodic interference peaks in the shifted frequency spectrum.

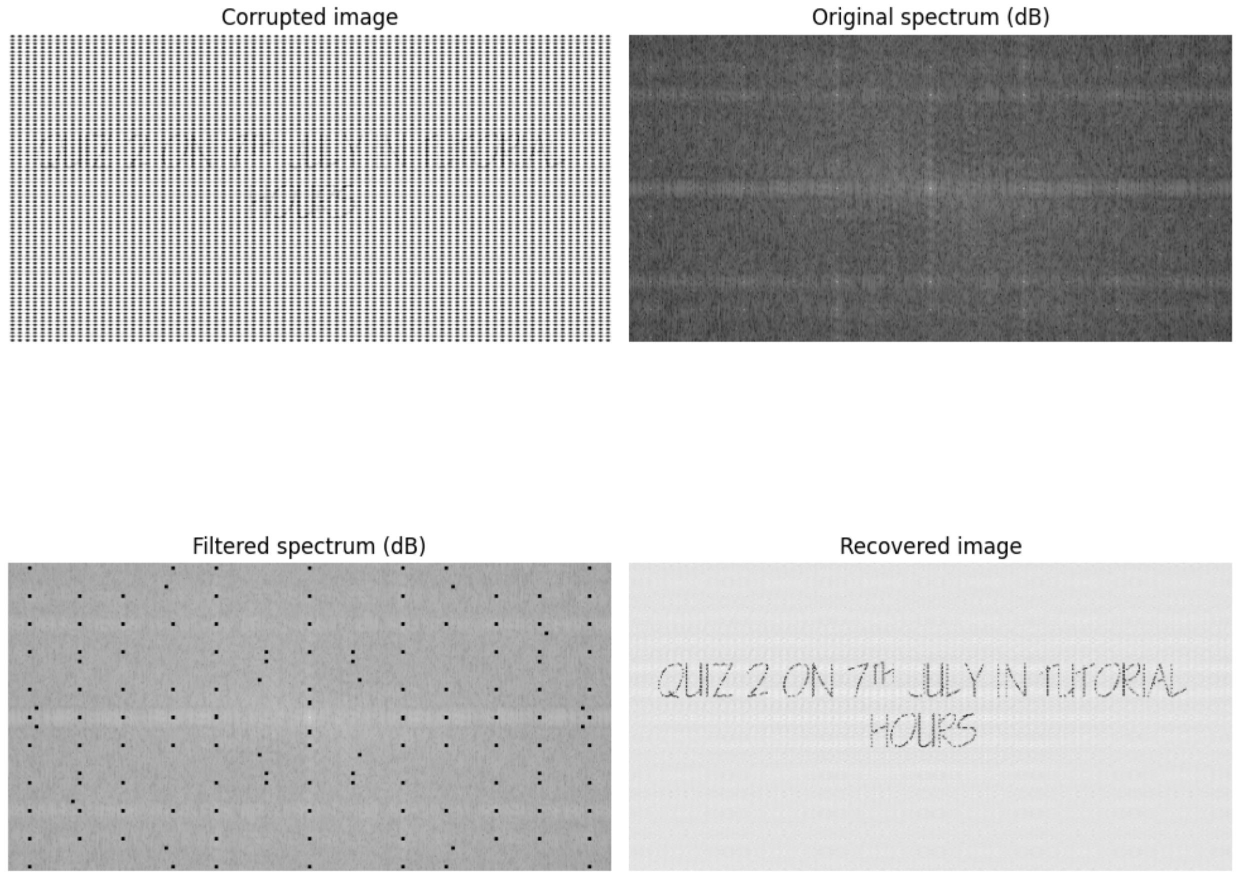
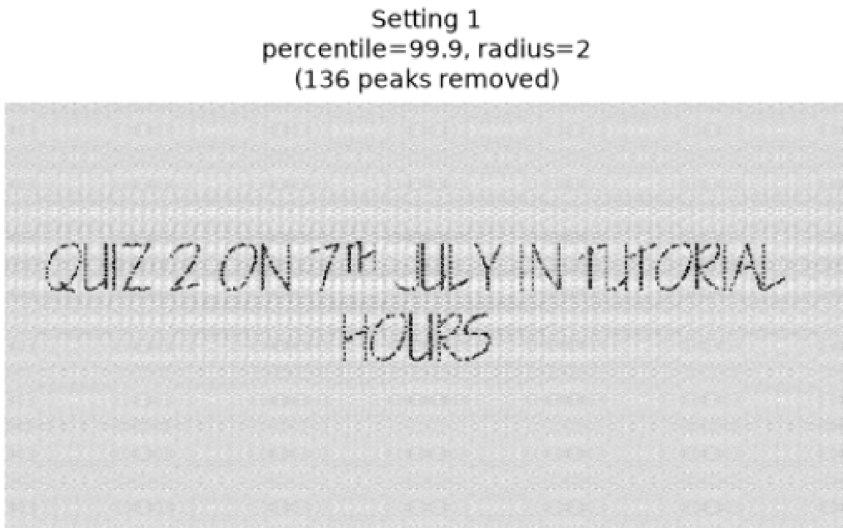
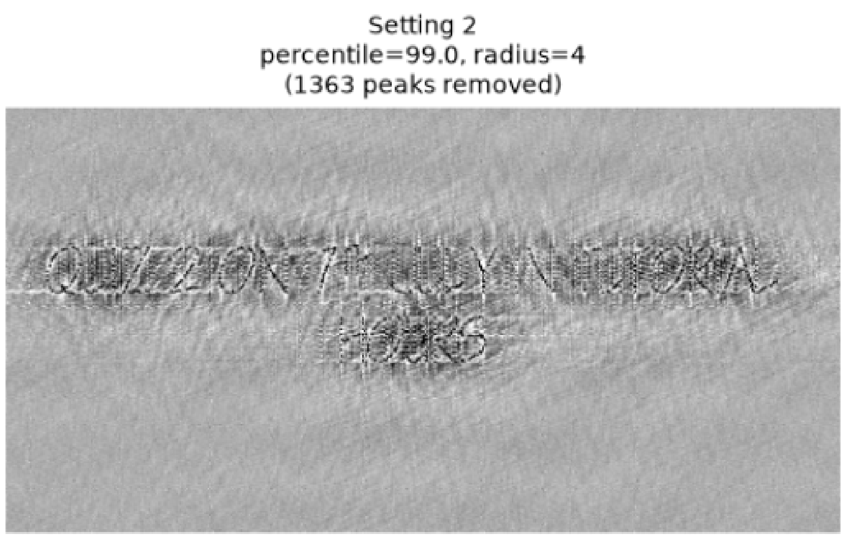


Figure 5: Corrupted image, original spectrum, filtered spectrum, and recovered image.

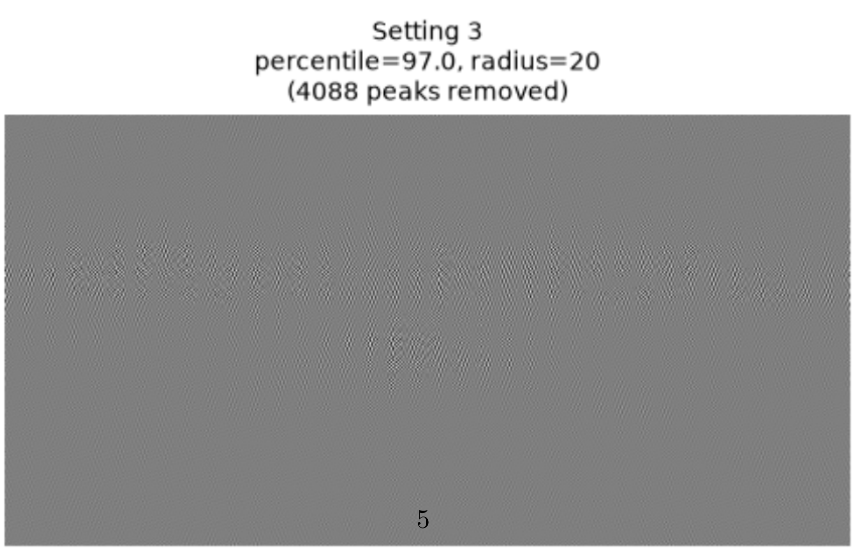
Figure 6: The recovered image is easy to read.



(a) Peak threshold = 99.9 percentile, notch radius = 2



(b) Peak threshold = 99.0 percentile, notch radius = 4



(c) Peak threshold = 97.0 percentile, notch radius = 20

Discussion

- **Was the message recovered?** Yes. The recovered message reads *"QUIZ 2 ON 7th JULY IN TUTORIAL HOURS"*.
- **Choice of Filter.** If the filter is too weak , then the corrupted frequency still remains in the reconstructed image and if the filter is too strong then useful image information is lost.
- **Does removing more frequencies always help? No.** It gives the same result as the filter if too much frequencies is remove useful information about the reconstruted image is also lost.
- **Applications.** Similar filtering techniques are used in photo-editing software to blur the background, in music and video apps to suppress unwanted noise and in noise cancellation devices.

Q1B. Digital Detective – "Missing Boundaries"

Objective

The objective is to identify object boundaries in the given image using the Sobel operator and to study how noise and image smoothing influence the detected edges.

Method

1. First we will load the given image and convert it to grayscale.
2. Then we will apply the Sobel operator in the horizontal and vertical directions using the kernels
$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \qquad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}.$$
3. Now to obtain the edge map which highlights the location where pixels intensity changes rapidly, we will compute the gradient magnitude
$$G = \sqrt{G_x^2 + G_y^2}$$
4. We then add Gaussian noise to simulate a noisy image and observe how noise affects Sobel edge detection.
5. Apply Gaussian smoothing with different values of σ before edge detection and compare the results.
6. Analyze the trade-off between noise reduction and preservation of weak edges.

Results

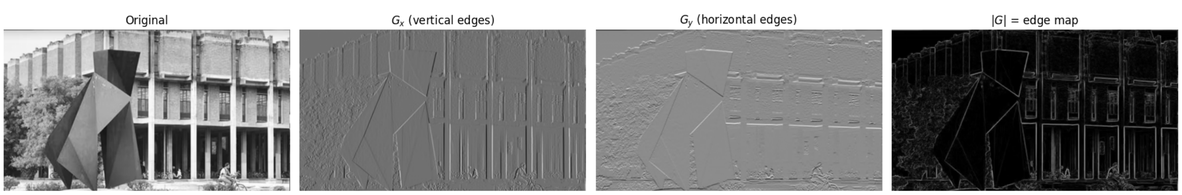


Figure 8: Original image, G_x, G_y, and the combined gradient magnitude $|G|$ obtained using the Sobel operator.

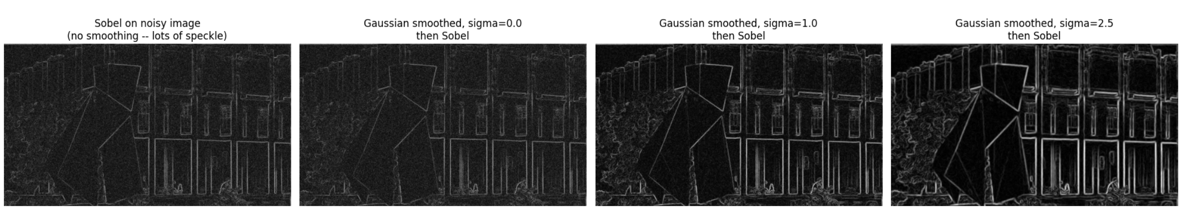


Figure 9: Effect of Gaussian smoothing on Sobel edge detection for different values of σ.

Discussion

- **Intensity changes** We will use derivation to identify the region where the intensity changes.
- **Derivatives and gradients.** Derivatives measure the rate of change of image intensity. Image gradient measures the rate of intensity change in both direction ie horizontal and vertcal. So larger value means edges or boundary and smaller means smooth region.
- **Sobel filter.** It is an operator used to detec the edge with the help of two 3×3 convolution kernels to estimate intensity changes in both horizontal and vertical directions. These kernels produce the gradient components G_x and G_y. Thus the gradient magnitude is:

  $$G = \sqrt{G_x^2 + G_y^2}$$

  is then used to identify edges.
  Large value of G means more intensity changes so it correspoond to boundries.
- **Effect of noise and role of smoothing.** Noise creates false intensity changes, which is be detected as false edges by the Sobel filter. Smoothing reduces these fluctuations before edge detection, resulting in cleaner and more reliable boundaries.
- **Preserving weak edges.** Weak edges can be preserved by using moderate Gaussian smoothing. Excessive smoothing removes weak edges, whereas insufficient smoothing allows noise to appear as false edges.